

A Computer-Assisted Classification of Knots Realizable with At Most Nine Sticks

The K9 Computer-Assisted Proof Project
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Abstract. We give a reproducible computer-assisted classification of ordinary unoriented knot types which admit a polygonal realization with at most nine straight edges (sticks), with mirrors identified. Subject to three explicit mathematical inputs - the complete rank-three oriented-matroid catalogue on eight elements, the standard low-crossing knot census, and Calvo's at-most-eight-stick classification - we prove that the set consists of 38 types. Subtracting Calvo's twelve at-most-eight types gives exactly 26 types whose stick number is nine. The proof combines a general-position roof reduction, finite source enumeration, strict diagram certificates, and exact rational-coordinate witnesses.

1. Statement and conventions

For a knot type K , let $s(K)$ be the least number of edges in an embedded closed polygon in $\mathbb{R}^3$ representing K , and write $K9=\{K:s(K)\leq 9\}$. Knots are ordinary, unoriented types and mirrors are identified. Let S be the 38 types listed below.

class	types
unknot	0_1
3-6 crossings	3_1, 4_1, 5_1, 5_2, 6_1, 6_2, 6_3
7 crossings	7_1,...,7_7
8 crossings	8_16, 8_17, 8_18, 8_19, 8_20, 8_21
9 crossings	9_29, 9_34, 9_35, 9_39,...,9_49
composite	square, granny, 3_1#4_1

Theorem. Under the external inputs and certificate checks described here, $K9=S$.

Calvo's Theorem 1(iv) gives twelve mirror-folded at-most-eight types: 0_1, 3_1, 4_1, 5_1, 5_2, 6_1, 6_2, 6_3, square, granny, 8_19, 8_20. Hence the exact-nine list is 7_1-7_7; 8_16, 8_17, 8_18, 8_21; 9_29, 9_34, 9_35, 9_39-9_49; and 3_1#4_1, for a total of 26 types.

2. Proof architecture

The proof is a pair of inclusions. Exact rational-coordinate witnesses establish $S \subset K9$. For the reverse inclusion, a nine-stick polygon is put in general position and reduced to a seven-crossing branch or a finite signed diagram corpus. Every corpus record is normalized under the global back-side operation, traversal reversal, relabelling, and mirror folding, then connected by a replayable certificate DAG to S . The gates are G1 source coverage, G2-G4 normalization and naming, G5 the 38 witnesses, and G6 the low-crossing and Calvo inputs.

3. Finite source reduction

After subdivision and perturbation, an exposed vertex becomes a roof under $F(x,y,z)=(x/z,y/z,1/z)$; the remaining seven edges carry finite affine heights. Radial projection gives a uniform rank-three oriented matroid on eight elements. The complete external catalogue has 135 classes. The replay enumerates 43,545,600 source contexts, retains 5,269,252 assignments after source-fibre union, and produces 83,023 signed records. Four-line and five-line identities are necessary height conditions. No conjectural global $a2$ bound is used.

4. Certificates and complete coverage

The global back-side view exchanges every over/under choice and preserves the ordinary knot type. It yields 52,347 representatives: 1,068 fixed flip orbits, 30,676 paired orbits, and 20,603 one-sided representatives. Strict RI, RII, RIII, and proper-strand-pickup certificates preserve darts, ports, cyclic orders, and parent hashes. The naming DAG has exact domain 83,023 and exact image 52,347; missing, duplicate, foreign, and non-S terminals are zero.

5. External inputs and limitations

The oriented-matroid catalogue, low-crossing census, standard DT naming database, and Calvo's theorem are external inputs. The offline package contains frozen sources, certificates, exact witnesses, a fixed Windows x64 runtime, and verify.ps1. The relocated replay returned PASS_PORTABLE_FULL_PROOF_CHAIN, with PASS_EXACT_COMPLETE_COVER for the graph stage and PASS_FRESH_EXACT_38_INCLUSIONS for witnesses. This is a computer-assisted theorem under the stated inputs, not a formal verification of the operating system, compiler, Python implementation, or cited classifications, and it does not replace a future purely structural proof.

References

Calvo, Geometric Knot Spaces and Polygonal Isotopy, arXiv:math/9904037v2, Theorem 1(iv). Finschi, Catalog of Oriented Matroids, uniform rank-three catalogue on eight elements. Menasco, Alternating Knots, arXiv:1901.00582v1.